COMPUTER SYSTEM VALIDATION AND QUALIFICATION PROCESS

SOP-006 | Standard Operating Procedure | Issued under POL-006

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Document Control

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Version	Date	Author	Description
0.1	[DATE]	[Author]	Initial draft
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1. Purpose

This SOP defines the end-to-end process for validating and qualifying GxP computer systems at [Company Name]. It operationalises POL-006 by establishing the mandatory lifecycle stages, deliverable requirements per validation track, Validation Master File (VMF) structure, and procedures for maintaining the validated state throughout a system's operational life.

Validation is not a one-time event — it is a continuous commitment. A system enters a validated state at the end of the validation lifecycle and must be actively maintained in that state through controlled change management, periodic review, and documented incident handling. This SOP governs all three phases: initial validation, ongoing maintenance, and eventual system retirement.

> **NOTE:** This SOP applies to all computer systems classified as GxP per SOP-005 / WI-005. The GxP flag — not the system tier — determines applicability. A Tier 3 system with a GxP flag is fully subject to this SOP. A Tier 1 system without a GxP flag is not. When in doubt, consult Compliance / QA.

2. Scope

This SOP applies to all computer systems used to create, process, store, modify, maintain, archive, retrieve, or transmit GxP records — including but not limited to: Laboratory Information Management Systems (LIMS), Electronic Batch Record systems (EBR), Manufacturing Execution Systems (MES), Chromatography Data Systems (CDS), Electronic Document Management Systems (EDMS) used for GxP documents, Clinical Trial Management Systems (CTMS), and GxP instrument control software.

This SOP also applies to infrastructure components (servers, network components, operating systems, database platforms) where they are directly in scope for the validated state of a GxP application — as determined during the risk assessment stage of this process.

The following are explicitly outside scope of this SOP: non-GxP computer systems (governed by POL-005 tier-based controls only); development and test environments not connected to production GxP data; manual paper-based records; and office productivity tools not used to create GxP records.

3. Roles and Responsibilities

Role	Responsibility in This Process
Compliance / QA Director	Process owner. Approves Validation Master Plan, Validation Report, and all major validation deliverables. Signs off validated state. Authorises GxP system go-live. Assesses regulatory notification obligations for validation deviations.
Data Custodian (IT)	Co-owner. Responsible for technical infrastructure, installation qualification (IQ) execution, system build documentation, and IT-side change management during validation activities.
System Owner	Business sponsor. Provides and approves User Requirements Specification (URS). Authorises go-live from a business process perspective. Confirms that the validated system meets business and regulatory requirements.
Validation Lead	Project-level role assigned per system. Coordinates validation activities, manages the validation schedule, maintains the VMF, and produces the Validation Report. May be from Compliance / QA or a qualified contractor.
Information Security	Reviews and approves security risk assessment inputs to the validation risk assessment. Confirms audit trail configuration and access control requirements are technically implemented.
DIO (Data Integrity Officer)	Reviews data integrity controls implemented during validation. Confirms ALCOA+ compliance for GxP data flows. Signs off DIRA update on system go-live.
Vendor / Supplier	For vendor-hosted or SaaS GxP systems: provides Supplier Quality Assessment documentation, audit evidence, change notification process, and Technical Agreement per POL-013.

4. Validation Track Selection

The first decision in any GxP system validation is track selection. POL-006 defines four assurance tracks (A–D) scaled to system complexity and regulatory risk. Track selection is made jointly by Compliance / QA and the System Owner at project initiation, documented in the Validation Master Plan, and may not be changed without a formal re-assessment.

Track	System Profile	Validation Approach	Key Deliverables
A — Full Validation	Custom-developed or highly configured systems directly processing GxP records. High complexity. Subject to 21 CFR Part 11 / EU Annex 11 in full.	Full GAMP 5-aligned lifecycle: URS → Functional Specification → Design Specification → Risk Assessment → IQ → OQ → PQ → Validation Report. Vendor audit required.	VMP, URS, FS, DS, Risk Assessment, IQ Protocol+Report, OQ Protocol+Report, PQ Protocol+Report, Traceability Matrix, Validation Report, User Training Records, System Release Note
B — Configured COTS	Commercial off-the-shelf system with significant configuration. Vendor leveraged for IQ; configuration-focused OQ/PQ.	Supplier Quality Assessment replaces source code review. Configuration specification drives OQ. PQ focuses on configured processes. Vendor audit preferred.	VMP, URS, SQA, Configuration Specification, Risk Assessment, IQ Protocol+Report, OQ Protocol+Report, PQ Protocol+Report, Validation Report
C — Standard COTS / SaaS	Standard commercial system with limited or no configuration. Vendor's validation documentation leveraged.	Vendor IQ documentation accepted with gap assessment. OQ covers system-specific configuration. PQ may be abbreviated. No source code review.	VMP, URS, SQA, Risk Assessment, OQ Protocol+Report, PQ Protocol+Report (may be combined), Validation Report
D — Infrastructure / Utility	Supporting infrastructure (OS, database, network) where GxP impact is indirect. Qualification rather than full validation.	Installation Qualification confirms correct installation per specification. Operational Qualification confirms performance requirements. No PQ required.	Qualification Plan, IQ Protocol+Report, OQ Protocol+Report, Qualification Summary Report

IMPORTANT: Track A systems: the Minor Change category defined in POL-007 does not apply. All changes to Track A systems, including those that would be minor elsewhere, require a Normal or Emergency change assessment and a validation impact assessment. This is a deliberate constraint

— the validated state of a Track A system is too consequential to admit informal changes.

5. Validation Lifecycle

The following stages apply to all validation tracks, with the depth of each stage scaled by track as defined in Section 4. All stages are mandatory — no stage may be skipped or retrospectively satisfied after system go-live without a formal deviation and Compliance / QA approval.

Step 01	**Project Initiation and Track Selection** Trigger: system identified as GxP via SOP-005 / WI-005, or material change to existing GxP system requiring re-validation. Actions: 1. System Owner raises a Validation Project Request with Compliance / QA 2. Compliance / QA and System Owner conduct initial system characterisation — regulatory context, intended use, data flows, user population 3. Track selected per Section 4; if ambiguous, Compliance / QA decides 4. Validation Lead assigned 5. Preliminary vendor assessment initiated per POL-013 / SOP-013 (parallel workstream) 6. VMF opened — all subsequent deliverables filed here Output: Track Selection Record filed in VMF.	Responsible **System Owner / Compliance / QA**
Step 02	**Validation Master Plan (VMP)** The VMP is the governing document for the entire validation project. It is approved by Compliance / QA before any testing activity begins. Mandatory VMP contents: — System description and intended use — Regulatory context and applicable regulations — Validation track and rationale — Validation scope boundary (what is and is not in scope) — Roles and responsibilities — Validation lifecycle stages and planned deliverables — Risk-based testing approach — Acceptance criteria philosophy — Deviation and change management approach within validation — Training requirements — VMF structure The VMP is a controlled document — any change during the validation project requires a formal VMP amendment approved by Compliance / QA.	Responsible **Validation Lead / Compliance / QA**
Step 03	**User Requirements Specification (URS)** The URS is the definitive statement of what the system must do to meet business and regulatory requirements. It is authored by the System Owner with support from subject matter experts and is approved by Compliance / QA before design or configuration activities begin. URS requirements must be: — Written in testable terms — 'the system shall...' not 'the system should...' — Uniquely numbered (URS-NNN format) for traceability — Regulatory requirements identified with their source citation (e.g., '21 CFR Part 11 §11.10(e)') — Inclusive of audit trail requirements, electronic signature requirements, access control requirements, and data integrity requirements for all GxP data flows The URS drives the Traceability Matrix — every URS requirement must be traceable to at	Responsible **System Owner / Validation Lead**

	least one test protocol entry.	
Step 04	**Risk Assessment** A system-level risk assessment is conducted to identify risks to data integrity, patient safety, and product quality — and to focus the validation testing effort on the highest-risk functions. This is not a security risk assessment (that is conducted separately per POL-010 / SOP-010), although inputs are shared. Risk assessment process: 1. Identify GxP data flows and critical system functions 2. For each function: assess the potential failure mode, the consequence for GxP records, and the probability of failure 3. Score severity (1–3) × probability (1–3) = risk score (1–9) 4. High-risk functions (score ≥6): full protocol testing required, no sampling 5. Medium-risk functions (score 3–5): protocol testing with defined sample size 6. Low-risk functions (score 1–2): verification or configuration confirmation acceptable 7. Risk assessment reviewed and approved by Compliance / QA The risk assessment drives protocol scope and is referenced in the Traceability Matrix.	Responsible **Validation Lead / Compliance / QA**
Step 05	**Installation Qualification (IQ)** IQ confirms that the system has been installed correctly in accordance with its specification. For vendor-hosted or SaaS systems, vendor IQ documentation is reviewed and a gap assessment performed. IQ verifies: — Hardware and infrastructure installation per specification — Software version confirmed against the approved specification — Configuration parameters documented — Security controls installed (access controls, MFA where required, audit trail enabled) — Network connectivity and integration points confirmed — Audit trail confirmed enabled and configured per POL-014 — NTP synchronisation confirmed and documented — Backup and recovery configuration documented IQ is executed by Data Custodian (IT) and approved by Compliance / QA. No OQ activities may commence until IQ is approved.	Responsible **Data Custodian (IT) / Compliance / QA**
Step 06	**Operational Qualification (OQ)** OQ confirms that the system operates as specified across its defined operational range — including edge cases, error handling, and security controls. Mandatory OQ test coverage for Track A/B systems: — All URS requirements mapped in Traceability Matrix — All high-risk functions (per risk assessment) fully tested — Audit trail validation: minimum 5 test points confirming timestamp accuracy, before/after capture, immutability, and user attribution — Audit trail cannot be disabled: test that no user role (including system administrator) can disable audit trail recording — Electronic signature validation (where applicable): signature binding, non-repudiation, re-authentication on signing — Access control validation: least privilege confirmed, MFA confirmed, session timeout confirmed — Error handling: system behaves correctly and records are protected on error conditions — Data backup and restore: confirmed that backup restores to complete and consistent state All OQ tests are executed by named, trained testers. All deviations (actual vs expected results) are documented as OQ deviations and resolved before PQ commences.	Responsible **Validation Lead / System Owner / Data Custodian (IT)**

	Unresolved Critical deviations block PQ execution.	
Step 07	**Performance Qualification (PQ)** PQ confirms that the system performs consistently as intended in the actual production environment with real users executing real processes. It is the final evidence-gathering stage before system release. PQ approach: — Executed by the actual system users (not the validation team) — this is a deliberate design: PQ tests both the system and the user training — Covers complete, representative end-to-end business processes including GxP record creation, modification, and approval workflows — Uses representative data volumes and production-representative configurations — Run in the production environment (or a production-identical environment) — not in a test sandbox — Includes audit trail review confirming that GxP records created during PQ are correctly captured, attributed, and immutable PQ acceptance criteria are defined in the PQ protocol and approved before execution. Unresolved Critical or High deviations at PQ completion block system release.	Responsible **System Owner / Business Users / Validation Lead**
Step 08	**Traceability Matrix and Validation Report** Before system release, the Validation Lead produces and Compliance / QA approves: Traceability Matrix: a document demonstrating that every URS requirement has been tested in at least one OQ or PQ test, and that every test result has been documented. Gaps in the matrix are findings that must be resolved before release. Validation Report: the definitive summary of the validation project. Mandatory contents: — Scope and purpose — Summary of lifecycle stages completed and deliverables produced — Summary of deviations and their resolution — Outstanding items (if any) and risk assessment for conditional release — Data integrity and audit trail confirmation — User training completion confirmation — Conclusion: system is / is not in a validated state. This statement must be unambiguous. — Approval signatures: Compliance / QA Director, System Owner, Data Custodian (IT) Conditional release — system released with known open items — requires Compliance / QA Director approval with documented risk acceptance and a binding remediation timeline. Open Critical or High items may not be carried over on a conditional release.	Responsible **Validation Lead / Compliance / QA Director / System Owner**
Step 09	**System Release and Go-Live** On Compliance / QA Director approval of the Validation Report: 1. System Release Note issued (version, date, validated scope, known open items) 2. GxP system activated in System Register (SOP-005) — GxP flag confirmed 3. DIRA Register updated (SOP-003) — system added as GxP system with ALCOA+ controls confirmed 4. Audit trail monitoring configured in SIEM per POL-014 (if not completed during IQ) 5. Access provisioned to named users per SOP-008 — no standing access during validation may carry forward without re-confirmation 6. VMF marked as In Validated State 7. Change Management controls activated per SOP-007 — all changes from this point require change management process No GxP activities may be conducted in the system until the System Release Note is issued.	Responsible **Compliance / QA Director / Data Custodian (IT) / System Owner**

6. Validated State Maintenance

Once a system is in a validated state, maintaining that state is an ongoing operational obligation. The validated state is not a historical achievement — it is a current status that can be lost at any moment through an uncontrolled change, a configuration drift, or an unresolved deviation.

6.1 Change Management

All changes to a GxP system in validated state are subject to SOP-007. The change management process determines the validation impact of each proposed change before it is implemented.

Change Type	Validation Impact Assessment Required	Deliverables Required	Approval
Minor Change (Track B/C/D only — Track A has no minor category)	Yes — Validation Lead confirms change falls within validated configuration boundary and does not alter GxP data flows, audit trail behaviour, or security controls	Change record (SOP-007). Validation Impact Assessment document. VMF change note.	Compliance / QA confirmation
Normal Change — no GxP impact	Yes — Validation Lead confirms change has no impact on validated state	Change record (SOP-007). Validation Impact Assessment document. VMF change note.	Compliance / QA confirmation
Normal Change — GxP impact (configuration or functional change)	Yes — full impact assessment determining re-validation scope	Change record (SOP-007). Validation Impact Assessment. Re-qualification protocol (scope per impact assessment). Updated Validation Report or Addendum. Updated Traceability Matrix.	Compliance / QA Director
Emergency Change on GxP system	Yes — immediate assessment. If validated state impact: Compliance / QA notified before implementation. If patient safety risk from not	Emergency change record (SOP-007). Immediate Validation Impact Assessment. Retrospective protocol execution within agreed timeline (maximum 10 business days for Track A).	Compliance / QA Director within 24hr

Change Type	Validation Impact Assessment Required	Deliverables Required	Approval
	implementing: verbal authorisation with 24-hour documentation.		
Major Upgrade requiring re-validation	Full re-validation — treated as new system validation for affected scope	VMP addendum or new VMP. Full IQ/OQ/PQ per track. New or updated Validation Report.	Compliance / QA Director

6.2 Periodic Review

All GxP systems in validated state are subject to periodic review confirming that the system continues to meet its validated requirements and remains in compliance with applicable regulations.

Review Type	Frequency	Scope	Output
Annual Validation Review	Annually per system	Review of: change history since last review; deviation log; audit trail review reports; access review; vendor audit status; regulatory change assessment; system performance against URS	Annual Review Report filed in VMF. Any findings that affect validated state status raised as deviations.
Regulatory Change Assessment	On publication of relevant regulatory guidance or on regulatory agency request	Assessment of whether new or amended regulatory guidance affects the validated state of the system or the validation approach	Regulatory Change Impact Assessment. VMF update. Re-qualification if required.
Audit Trail Review	Per SOP-014 cadence (GxP systems: weekly minimum)	Audit trail records reviewed for completeness, integrity, and anomalies per SOP-014 and SOP-003	Audit Trail Review Record (GxP quality record — retained per POL-002). Anomalies escalated per SOP-003.

6.3 Deviation Management

A validation deviation is any departure from an approved validation protocol, procedure, or specification — detected either during validation activities or during operation of the validated system. All deviations are documented, investigated, and resolved.

Deviation Category	Examples	Response	Impact on Validated State
Critical	Audit trail found to be modifiable; GxP data found to be alterable without audit trail capture; system found to be operating outside validated configuration	Immediate Compliance / QA notification. System placed in quarantine (no GxP activities) pending investigation. Root cause analysis. CAPA. Retrospective validation of affected records if possible.	Validated state suspended until deviation resolved and Compliance / QA Director confirms restoration
Major	OQ or PQ test failure affecting GxP record integrity; access control misconfiguration on GxP system; configuration drift from validated baseline	Compliance / QA notification within 24 hours. Investigation and root cause. CAPA. Re-testing of affected protocol sections.	Validated state reviewed — may continue under enhanced monitoring pending CAPA completion. GxP activities restricted to unaffected scope where possible.
Minor	Documentation error in validation protocol; non-GxP configuration parameter outside specification; user training gap (user trained after access granted)	Documented. Root cause. Corrective action within 30 days.	No impact on validated state — deviation does not affect GxP data integrity or regulatory compliance

6.4 Operational Incident Impact

When an operational incident (per POL-015 / SOP-015) affects a GxP system, the validation impact must be assessed before GxP activities resume. The SOP-015 Major Incident process includes mandatory Compliance / QA notification for all GxP system incidents. The following applies on resolution:

- System restored to operation: Data Custodian (IT) confirms system is running. This is not a validated state confirmation.
- Validated state assessment: Compliance / QA and Validation Lead assess whether the incident or its resolution (e.g., emergency patch, configuration restore from backup) affects the validated state.
- Audit trail review: audit trail reviewed for the incident period to confirm completeness and integrity per SOP-003. Gaps are ALCOA+ Completeness failures and handled as Critical deviations.
- GxP restart authorisation: Compliance / QA Director signs off that the system is in validated state before any GxP activities resume. This sign-off is documented in the VMF as an incident record.

7. System Retirement and Data Migration

System retirement from validated state requires a controlled process to ensure that GxP records are preserved and accessible for their full required retention period, and that the validated state is formally closed.

Step 01	**Retirement Planning** System Owner and Compliance / QA develop a retirement plan covering: — GxP record inventory: all records held in the system, their retention periods (per POL-002), and their regulatory context — Data migration or archival approach: records migrated to a replacement system (which must itself be validated) or archived in a compliant long-term archive — Access continuity: how GxP records will be accessible in human-readable format for inspectors for the full retention period — Audit trail preservation: audit trail records preserved alongside the records they protect — Read-only period: if records cannot be immediately migrated, the system enters a read-only validated state until migration is complete	Responsible **System Owner / Compliance / QA**
Step 02	**Data Migration Validation** If GxP records are migrated to another system: — Migration is treated as a validation activity — the migration itself must be validated — A migration validation protocol is produced and approved by Compliance / QA before migration begins — Migration validation confirms: all records migrated (completeness), records are identical to source (accuracy), audit trail records migrated alongside data records (integrity), and the receiving system correctly displays records in human-readable format — A 100% record count reconciliation is mandatory; sampling is not acceptable for GxP record migration — Migration Validation Report approved by Compliance / QA Director before source system is decommissioned	Responsible **Validation Lead / Compliance / QA**
Step 03	**Validation Closure Report** On completion of retirement: — Validation Closure Report produced summarising: retirement rationale, data migration/archival outcome, record retention confirmation, audit trail disposition — Approved by Compliance / QA Director and System Owner — VMF closed and archived — retained for the life of the GxP records the system held plus the applicable GxP retention period — System Register (SOP-005) updated: GxP flag removed, status set to Retired — DIRA Register (SOP-003) updated: system removed from active GxP data inventory — Vendor offboarding per SOP-013: Validation Closure Report shared with vendor; access revoked; Technical Agreement formally closed	Responsible **Compliance / QA Director / Validation Lead**

8. Validation Master File (VMF) Structure

Each GxP system maintains a VMF — the complete, indexed record of all validation activities for the system. The VMF is a GxP quality record and must be maintained with the same controls as any other GxP record: version-controlled, protected from unauthorised modification, and retained for the full required period.

VMF Section	Contents
1. System Overview	System description, intended use, regulatory context, validation track, VMF owner and custodian
2. Project Initiation	Track Selection Record, project team, VMP
3. Regulatory and Risk	Regulatory context document, system risk assessment, security risk assessment inputs
4. Requirements	URS, Functional Specification (Track A/B), Design Specification (Track A)
5. Vendor Assessment	Supplier Quality Assessment, Technical Agreement, audit reports, certification copies (ISO 27001, SOC 2)
6. Qualification Protocols and Reports	IQ, OQ, and PQ protocols (approved versions), execution records, deviation logs, protocol reports
7. Traceability Matrix	Current approved version, version history
8. Validation Report	Current approved version; addenda for subsequent re-validations or change-driven qualification
9. System Release	System Release Notes, go-live confirmation, DIRA update record
10. Change History	All change records (per SOP-007) affecting the validated state, with Validation Impact Assessments
11. Deviation and CAPA Log	All deviations and associated CAPAs — status tracked to closure
12. Periodic Reviews	Annual Review Reports, regulatory change impact assessments
13. Operational Incident Records	Incident records (per SOP-015) that affected the validated state, with validated state restoration sign-offs
14. Audit Trail Review Records	GxP audit trail review records (per SOP-014), anomaly reports
15. Training Records	Training completion records for all GxP system users — indexed by user and training event
16. Retirement	Retirement Plan, Data Migration Validation Report, Validation Closure Report

9. Related Documents

Document ID	Title	Relationship
POL-006	Computer System Validation and Qualification Policy	Parent policy — this SOP operationalises POL-006
POL-003	Data Integrity Policy	ALCOA+ data integrity requirements are the core regulatory driver for GxP validation; audit trail validation tests ALCOA+ Attributable, Legible, Contemporaneous, and Complete
POL-005	System Classification Policy	GxP flag assigned during system classification (SOP-005/WI-005) is the trigger for this SOP
POL-007	Change Management Policy	All changes to GxP systems in validated state are governed by SOP-007; change management is the primary mechanism for maintaining validated state
POL-009	Privileged Access Management Policy	Privileged access to GxP systems is governed by POL-009/SOP-009; privileged access configurations are validated in IQ/OQ
POL-010	Information Security Risk Policy	Security risk assessment inputs shared with validation risk assessment; vulnerability management for GxP systems must be sequenced with validation impact assessment
POL-011	Business Continuity Policy	DR provisions for GxP systems must be validated in OQ; validated state restoration is a mandatory element of GxP DR recovery per SOP-011
POL-013	Third Party and Vendor Risk Policy	GxP vendor qualification (5-step process) runs in parallel with validation — vendor Technical Agreement and SQA are VMF deliverables
POL-014	Audit Trail and Monitoring Policy	Audit trail configuration is validated in IQ; audit trail immutability and cannot-be-disabled requirements are tested in OQ; ongoing audit trail review per SOP-014
POL-015	IT Service Management Policy	Operational incidents on GxP systems require validated state assessment and Compliance/QA sign-off before GxP restart per this SOP
SOP-003	Data Integrity Process	DIRA Register updated on GxP system go-live; audit trail anomalies detected in periodic review trigger SOP-003 process
SOP-005	System Classification Process	System classification and GxP flag assignment precede validation initiation; System Register updated on validated state go-live
SOP-007	Change Management Process	All validated state changes managed through SOP-007 with validation impact assessment per Section 6.1 of this SOP
SOP-013	Third Party and Vendor Risk	GxP vendor qualification coordinated between

Document ID	Title	Relationship
	Assessment Process	this SOP and SOP-013; vendor audit evidence filed in VMF
SOP-014	Audit Trail Review and Monitoring Process	Routine GxP audit trail review per SOP-014 is a validated state maintenance activity
EXC-001	Exception Management Policy and Register	GxP validation exceptions (e.g., overdue re-qualification) are Category 4 under EXC-001 with 6-month maximum and mandatory Compliance/QA involvement; operating an unvalidated GxP system is non-approvable
WI-005	System Classification Questionnaire	Regulatory flag determination in WI-005 is the primary trigger for validation initiation under this SOP
RACI-001	Roles, Responsibilities and Authorities	Accountability for validation governance

10. Definitions

Term	Definition
ALCOA+	Data integrity framework: Attributable, Legible, Contemporaneous, Original, Accurate — plus Complete, Consistent, Enduring, Available
GxP	Good Practice — collective term for pharmaceutical quality guidelines including GMP, GLP, GCP, GDP
IQ	Installation Qualification — documented verification that a system has been installed correctly in accordance with its specification
OQ	Operational Qualification — documented verification that a system operates as specified across its defined operational range
PQ	Performance Qualification — documented verification that a system consistently performs as intended in the production environment
URS	User Requirements Specification — the definitive statement of what the system must do to meet business and regulatory requirements
Validated State	The condition of a computer system that has been validated according to an approved Validation Master Plan and is operating within its validated specification
VMF	Validation Master File — the complete, indexed record of all validation activities for a system, maintained as a GxP quality record for the life of the system plus the applicable retention period
VMP	Validation Master Plan — the governing document for a validation project, approved by Compliance / QA before testing begins

Term	Definition
CAPA	Corrective and Preventive Action — the documented process for investigating the root cause of a deviation and implementing corrective actions to prevent recurrence
SQA	Supplier Quality Assessment — the documented evaluation of a vendor's quality management system, validation practices, and GxP compliance

— End of Document —

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